A LOCALLY ACYCLIC MORPHISM IS LOCALLY ACYCLIC RELATIVE TO EVERY LOCALLY CONSTANT TORSION SHEAF OF PRIME-TO-CHARACTERISTIC ORDER

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file etale-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the proof is written out in full: the trivialisation at a geometric point, the filtered-colimit decomposition of the stalk into finite prime-to-characteristic groups and Z/nZ summands, and the computation of the cohomology of the affine Milnor fibres with the colimit lemma.

Editorial note. The label lemma-locally-acyclic-locally-constant names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Editorial note. The statement and proof are copied from the pinned chapter file; the two cited inputs are kept as external pointers with their canonical Stacks tags.

Editorial note. The preamble is the chapter's own native preamble excerpt already used for this source: the Xy-pic setup (native preamble lines 14-18) and the native theorem declarations with the mathematical macros (native preamble lines 154-217).

Allowed standard prerequisites (editorial).

- Local acyclicity of f and its description through the fibres F_{x,t}. Same chapter, the definition of local acyclicity and the discussion following it, giving the vanishing of the cohomology of the constant sheaves Z/nZ on the affine Milnor fibres. Supplies the hypothesis on f and the vanishing computation used for each Z/n_{i,j}Z.
- Constant sheaves and filtered colimits of cohomology. Same chapter, the lemma computing the cohomology of a constant sheaf with coefficients in a filtered colimit as the colimit of the cohomologies. Supplies the identity H^q(F_{x,t}, underline M) = colim H^q(F_{x,t}, underline M_i) for the finite subgroups of M.

Source reference. Original reference 0GJP: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 19110–19110. Prerequisite pointer invoked at proof line(s) 19274; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 00D5: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 7427–7427. Prerequisite pointer invoked at proof

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line(s) 19259; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let $f : X \to S$ be a morphism of schemes. Let $\mathcal{F}$ be a locally constant abelian sheaf on $X_{étale}$ such that for every geometric point $\overline{x}$ of X the abelian group $\mathcal{F}_{\overline{x}}$ is a torsion group all of whose elements have order prime to the characteristic of the residue field of $\overline{x}$. If f is locally acyclic, then f is locally acyclic relative to $\mathcal{F}$.*

Proof. Namely, let $\overline{x}$ be a geometric point of X. Since $\mathcal{F}$ is locally constant we see that the restriction of $\mathcal{F}$ to $\mathrm{Spec}(\mathcal{O}^{sh}_{X,\overline{x}})$ is isomorphic to the constant sheaf $\underline{M}$ with $M = \mathcal{F}_{\overline{x}}$. By assumption we can write $M = \mathrm{colim}\, M_i$ as a filtered colimit of finite abelian groups M_i of order prime to the characteristic of the residue field of $\overline{x}$. Consider a geometric point $\overline{t}$ of $\mathrm{Spec}(\mathcal{O}^{sh}_{S,f(\overline{x})})$. Since $F_{\overline{x},\overline{t}}$ is affine, we have

$$H^q(F_{\overline{x},\overline{t}}, \underline{M}) = \mathrm{colim}\, H^q(F_{\overline{x},\overline{t}}, \underline{M_i})$$

by Lemma 00D5. For each i we can write $M_i = \bigoplus \mathbf{Z}/n_{i,j}\mathbf{Z}$ as a finite direct sum for some integers $n_{i,j}$ prime to the characteristic of the residue field of $\overline{x}$. Since f is locally acyclic we see that

$$H^q(F_{\overline{x},\overline{t}}, \underline{\mathbf{Z}/n_{i,j}\mathbf{Z}}) = \begin{cases} \mathbf{Z}/n_{i,j}\mathbf{Z} & \text{if} \quad q = 0 \\ 0 & \text{if} \quad q \neq 0 \end{cases}$$

See discussion following Definition 0GJP. Taking the direct sums and the colimit we conclude that

$$H^q(F_{\overline{x},\overline{t}}, \underline{M}) = \begin{cases} M & \text{if} \quad q = 0 \\ 0 & \text{if} \quad q \neq 0 \end{cases}$$

and we win. □